

Omar AbdelKader

Doctoral Researcher in Computer Science
Inria - EVREF / LaBRI - Progress
University of Lille
Lille, France

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omar.abedekader@inria.fr
omarabedekader.com
github.com/omarabedekader
ORCID 0009-0005-1339-5683

About Me

I am an AI researcher with extensive experience in artificial intelligence and software engineering.
I am currently pursuing a Ph.D. in Computer Science at the University of Lille.

Responsibilities

Leadership and Founding Roles

- Founder and Maintainer of the organization **pharo-llm** [🔗](#) on github and **pharo-llm** [🔗](#) on hugging face.
- President of **Synapse-NeuroTech-Lille** [🔗](#), a student association focused on AI and neuroscience.
- Project Leader and Maintainer for multiples projects **chatpharo** [🔗](#), **pharo-infer** [🔗](#) and many **more**.

Community and Organizational Service

- Member of the organizing Committee, **GDR-SciLog'26** [🔗](#).
- Member of the organizing Committee, **ESUG'26** [🔗](#), **ESUG'25** [🔗](#), **ESUG'24** [🔗](#).
- Member in the organization **pharo-ai** [🔗](#).
- Member in **AI and Ecological Transition Club** [🔗](#) at **Ministry of Ecological Transition** [🔗](#).
- Contributor in **Café IA** [🔗](#) at **National Digital Council** [🔗](#).

Reviewing, Mentoring, and Academic Supervision

- Sub reviewer at **BENEVOL'25** [🔗](#).

Awards and Honors

Awards:

- Best Paper Award (3rd place) — IWSST 2025, Gdańsk, Poland
“Package-Aware Approach for Repository-Level Code Completion in Pharo” [🔗](#)

Honors:

- Alumni Testimonial — IDMC 2024, University of Lorraine, France
“Alumni testimonial in NLP & AI in University of Lorraine” [🔗](#)

Publications

Rank A

- Kier, K., Giagnorio, A., Abdelkader, O., Zaitsev, O., Peharz, R., Robbes, R., Bavota, G., & Ducasse, S. (2026). Teaching LLMs a Low-Resource Language: Enhancing Code Completion in Pharo. *International Conference on Software Maintenance and Evolution (ICSME)*. DOI.

Rank C

- AbdelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2025). Package-Aware Approach for Repository-Level Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWSST)*. DOI.
- Elkolei, M., AbdelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWSST)*. DOI.

Q2 Journals

- AbdelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2026). Evaluating Package-Level Scoping Strategies for Repository-Level Code Completion in Pharo. *Journal of Computer Languages (COLA)*. DOI.

Experience

PhD Student — INRIA, [EVREF], & Labri [Progress] (Villeneuve-d'Ascq, France) Oct. 2024 – Present

- Improving code completion and code generation using LLMs, specifically targeting the Pharo programming language, which has limited training data.
- Developing techniques for code completion, type inference, and deployment in Pharo's IDE, with a focus on runtime performance.

AI Engineer — INERIS, [CIVIS] (Verneuil-en-Halatte, France) Sept. 2023 – Sept. 2024

- Developed “INERIS-IA,” a tool to classify textual documents according to INERIS's strategic goals using ML and NLP techniques.
- Created Boolean queries for document retrieval and improved corpus quality through document similarity and keyword extraction.

Intern — LIPN, [RCLN] (Villetaneuse, France) June 2023 – Aug. 2023

- Compared sampling techniques for probabilistic planning, particularly for generating literary narratives.
- Evaluated methods including the Score Function Estimator (SFE) and advanced techniques such as Gumbel-Softmax, assessing their effectiveness in producing coherent and creative stories.

Software Developer — BEON-IT (Beirut, Lebanon) May 2022 – June 2022

- Java · Design Patterns · .NET Framework · Threading · Microsoft SQL Server

Full-Stack Developer — SSCC-IT (Andket, Lebanon) Nov. 2020 – Apr. 2021

- PHP & SQL · Design Patterns · Microsoft SQL Server
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Education

University of Lille — Villeneuve-d'Ascq, France

- Ph.D. in Computer Science (Oct. 2024 – Present)

University of Lorraine — Nancy, France

- M.Sc. in Natural Language Processing (Sept. 2022 – Sept. 2024)

Lebanese University — Beirut, Lebanon

- B.Sc. in Data Science (Sept. 2019 – July 2022)
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Software

ChatPharo 

- Live conversational interface between Pharo developers and large language models.
- Enables interactive discussions with LLMs directly from the Pharo environment.
- Designed an open and extensible architecture to experiment with multiple LLM backends.

INERIS-IA 

- AI platform developed at INERIS for document classification and knowledge management.
- Built a Flask-based web interface integrating multiple ML and NLP models.
- Trained document classification models aligned with COP 2027 objectives and INERIS strategic themes.
- Implemented additional features including document similarity, keyword extraction, and corpus quality improvement.

Family Dynamics Analysis  

- This project analyzes family dynamics in France using Formal Concept Analysis (FCA).
- Using data from the French national census, the study identifies patterns in family structures, single-parent households, child-bearing decisions, and naturalization through marriage across five regional zones.
- The project explores relationships between demographic variables such as marital status, household composition, and nationality, offering insights into cultural and social trends shaping French households.

Real-Fake Face Detection 

- Demonstrates the process of building and training a neural network for classification tasks.
- Walks through data preprocessing, model architecture design, training, evaluation, and visualization of results to assess performance and prediction accuracy.

EESMACF  

- Evaluates two NLP models—SBERT (Sentence-BERT) and MiniLM—for classifying analogies using the FrameNet dataset.
- The objective is to identify valid and invalid analogies by leveraging SBERT embeddings and a fine-tuned MiniLM classifier.
- MiniLM achieved 99% accuracy in distinguishing valid from invalid analogies, outperforming SBERT (~55% accuracy).

DeGatto  

- Sentiment analysis framework for e-commerce women's apparel reviews.
 - Uses a Kaggle dataset (23,000+ sentences) with aspect-level annotations for material, size, design, and comfort.
 - Evaluated multiple NLP/DL/ML models (BiLSTM, SVM, Logistic Regression, Multinomial Naive Bayes). BiLSTM performed best for sentence-level analysis, and LinearSVC performed well for aspect-level analysis.
 - Includes a visualization tool built with ReactJS and NodeJS to display results as bar or pie charts.
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Teaching

- Teaching assistant at IUT: *Introduction to Software Development* 41,5 h Labs 
 - Teaching assistant at IUT : *Software Maintenance* 12h Tutorials 
 - Teaching assistant at Polytech: *DataBase* 16h Labs 
 - Teaching assistant at Polytech: *Internet* 12h Labs 
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Supervision

- Mehdi Elkolei, Bachelor Internship, 4 month, April 2026 - July 2026
 - Srajal Dwivedi, Google Summer of Code, 3 month, May 2025 - July 2025
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Public Talks

- **French Brain Awareness Week 2026:** AI and the brain: two ways of learning, what does that change for our daily lives? [!\[\]\(0551a83d441798e532995956b603f604_img.jpg\)](#)
 - **ESUG 2025:** ChatPharo: An Open Architecture for Understanding How to Talk Live to LLMs [!\[\]\(54ee180c0037b66a36ce2219a481afde_img.jpg\)](#) [!\[\]\(73ae654e8897db9b21f1bf9d9efc07ef_img.jpg\)](#)
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Technologies & Skills

- **Programming Languages:** Python, Pharo, Java, JavaScript
 - **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Keras, Scikit-learn
 - **Natural Language Processing (NLP):** spaCy, NLTK
 - **Big Data & Distributed Systems:** Apache Spark, Hadoop
 - **Data Analysis & Visualization:** Pandas, Matplotlib, Seaborn
 - **Scientific Computing:** NumPy, SciPy
 - **Web Development:** Flask, Django, Laravel, ReactJS
 - **Databases & Cloud:** MongoDB, MySQL, AWS
 - **Operating Systems:** macOS, UNIX, Windows
 - **Research Areas:** Artificial Intelligence, Software Engineering, AI4SE, SE4AI
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Volunteering

President of Synapse-NeuroTech-Lille-ULille - (2024–Present) [!\[\]\(e474458956c9a37fbf9586ddb60a7fa1_img.jpg\)](#)

- Led the executive board and coordinated association activities and projects.
- Managed the annual budget, including financial planning, expense tracking, and funding allocation.
- Represented the association in meetings with university administration, partners, and sponsors.
- Oversaw event planning, workshops, and conferences related to neurotechnology and artificial intelligence.
- Defined strategic goals and ensured alignment with the association's mission.

Member in Club IA et Transition écologique at Ministère de la Transition Écologique — (2023–Present) [!\[\]\(5361750c22c4e047a52f4eac1ec2d4cc_img.jpg\)](#)

- Contributed to ongoing AI-driven projects supporting ecological transition initiatives within the French government.
- Assisted in the development and evaluation of AI use cases for public policy and environmental impact.
- Collaborated with interdisciplinary teams, including policymakers, researchers, and technical experts.
- Participated in strategic discussions on responsible and sustainable AI deployment in the public sector.

Contributor in Café IA at Conseil National du Numérique (CNNum) — (2023–Present) [!\[\]\(4fe57c3593bf1b21d272ae7ac8dfaf77_img.jpg\)](#)

- Contributed to national discussions on artificial intelligence, digital policy, and societal impact.
- Assisted in the preparation and facilitation of Café IA sessions bringing together experts, researchers, and public stakeholders.
- Participated in the analysis of AI trends, use cases, and regulatory challenges.
- Supported knowledge-sharing initiatives on responsible and ethical AI.

Volunteer First Aider in Croix-Rouge Française — (2022–2023) [!\[\]\(b792654f2cef9719eabeb6c5be00811e_img.jpg\)](#)

- Provided first aid and emergency assistance during public events and operations.
 - Responded to medical emergencies following established safety and emergency protocols.
 - Collaborated with multidisciplinary teams under high-pressure situations.
 - Maintained readiness through regular training in first aid and emergency response.
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Hobbies

- **Water Activities:** Swimming · Diving
 - **Outdoor and Adventure Sports:** Hiking · Skiing · Skydiving
 - **Creative and Artistic Activities:** Photography
 - **Precision Sports:** Boxing · Bowling · Golf
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Languages

- **Arabic:** Native
 - **French:** Native
 - **English:** B2
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